

Learning Gait Motion Without Labels: A Single-Stream Masked-JEPA Feasibility Study

CoDy-JEPA project · run 91108 · Health&Gait silhouettes (subject-disjoint) · NVIDIA H100

Summary. Labeled clinical gait video is scarce, so this experiment tests whether a model can teach itself walking motion from unlabeled silhouette video via **self-supervised learning**: the model hides part of each clip and learns to predict the hidden part from the rest. The run confirms the training signal is real, escapes the representational collapse that sank two earlier runs, and validates the full pipeline needed to later fine-tune backbones such as **V-JEPA 2** and **SAM 3D** for next-action prediction and balance-score estimation.

Method: predict hidden motion in feature space, not pixels

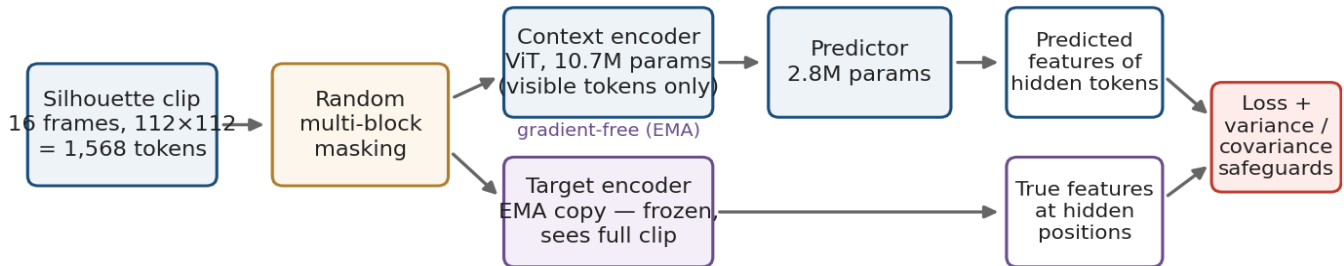


Figure 1 — A Joint-Embedding Predictive Architecture (JEPA). The context encoder sees only visible tokens; the predictor guesses the features of hidden tokens; a slowly-updated (EMA) target encoder supplies the answer. Predicting features rather than pixels focuses capacity on motion structure. Because a model can “cheat” by making all features identical (**collapse**), this run adds variance and covariance penalties plus target standardization — introduced after runs 90881/91023 collapsed to rank 2 with EMA alone.

Trained on 2,506 sequences from 318 subjects; validated on 624 sequences from 80 **different** subjects, so scores reflect generalization to unseen people, not memorization. 100 epochs, 3,900 steps, effective batch 64, ~24M parameters, bfloat16.

Results: real learning, no collapse — with one honest gap

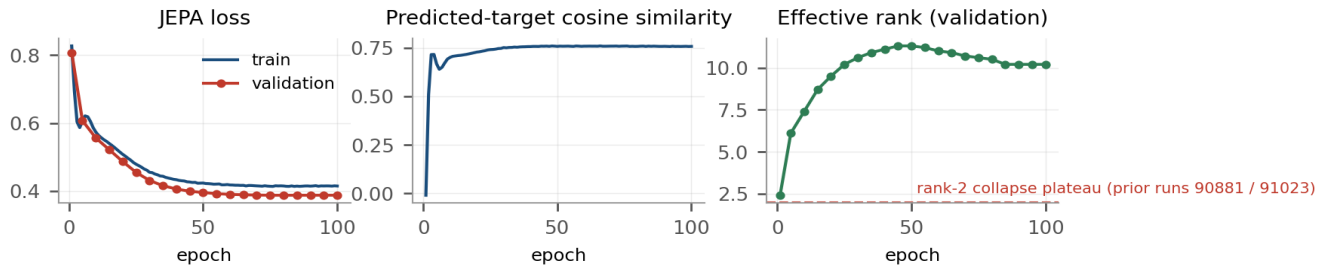


Figure 2 — Validation loss falls from 0.81 to a best of **0.387** (epoch 80) with no overfitting; cosine similarity between predicted and true hidden features rises from ≈ 0 to **0.76**; effective rank grows from 2.4 to a peak of **11.3**, a five-fold escape from the prior collapse plateau.

The automated health gate reports one unresolved limitation: shuffling clips so the model sees the *wrong* person’s context barely hurts prediction (gap ≈ 0.0002 vs. the required 0.001), and effective rank remains below its target ratio (0.027 vs. 0.05). The model avoids collapse but does not yet rely strongly on temporal context — the crux of learning *motion* rather than shape.

Why this matters for ambient intelligence and biomechanics

Ambient systems — passive monitoring in homes, clinics, and care settings — cannot depend on hand-labeled footage; a JEPA pretrained on abundant unlabeled video needs only small labeled sets for downstream adaptation, and the silhouette-first, identity-suppressing design fits ambient privacy expectations. For biomechanics, next-action prediction and balance scoring are inherently predictive and temporal: a model already trained to anticipate hidden future motion is the natural starting point, and the wrong-context diagnostic gives a principled test of whether a fine-tuned model learned genuine motion dependence — the difference between a balance score that transfers to new patients and one that does not.

Next steps

Close the context-dependence gap by strengthening the temporal objective and masking strategy; raise effective rank via safeguard tuning; run the frozen-feature probes (identity, held-out retrieval, subject-disjoint gait classification) to quantify what the features carry; then build toward full CoDy-JEPA — dual attribute/dynamics streams with counterfactual token swapping — and fine-tune V-JEPA 2 and SAM 3D on this validated pipeline.